

Presenting global water balance and consumption in the year 2019: Data cleaning, transformation and visualization

1. Introduction

Question: Propose a social science research question you plan to answer with your data.

RQ: *How do trends of water supply and use relate to water balance and consumption across different economic activities in various countries?*

At the beginning of the assignment, the research question addressed water scarcity instead of water balance and consumption. Damkjaer and Taylor (2017) describe in their paper the diverse ways to measure water scarcity developed throughout the years. Although this assignment was meant to measure the previous; it was not an easy task as some of these measurements required very detailed additional information, or their measurements were deemed to be outdated. Thus, the chosen dataset and the proposed research question may address patterns of water consumption and tentatively serve as a starting point for future measurement of water scarcity across the different economic sectors and within countries. Furthermore, the chosen data may manage to answer the research questions through visualizations.

2. Data

Question: Describe your data.

The data selected for answering the research question was taken from the OECD data explorer website (OECD, 2024). The name of the downloadable dataset in the webpage is “*Water accounts - supply and use*”, as its name describes, it holds information of the supply and use of water. The data was published on March 5 of 2024. It is not directly collected from the OECD but from different sources which are then sorted into a comprehensible dataset. The sources for European and OECD countries are from the OECD/Eurostat questionnaire on Inland Waters, and the source for non-European non-OECD countries is collected from UNSD data collection on environment statistics. The dataset downloaded was renamed as “oecd.csv” for simplicity and easier handling. Its total raw number of observations amounts to ($n = 37\,259$), and its columns/variables are ($n=28$). The variables of interest by broadly looking at the data at a first glance are 7 of which two are numeric and the rest are currently coded as characters. The numeric variables describe the year (1), and the water value (2) respectively. For the character variables they describe the following: country code (3), country name (4), type of water used/supplied (5), economic activity/sector (6), and supply/use of water (7).

3. Methodology

Question: Describe how you preprocessed your data and why.

The methodology employed for cleaning and processing the data had two steps: exploration (1), and cleaning (2). In the first step the data is explored to further understand what the dataset contains and what needs to be done for the cleaning process. In the second step the data is cleaned and transformed into tidy formats. **Exploration:** In order to check data as a whole, functions that can visualize and describe the structure of the data set are “str()”, and “unique()”. The first one functions to see which kind of variables the dataset has (numeric, factor, characters). The second function lets us see the unique values of each variable. This function is primordial as knowing the specific content of each variable makes it easiest to understand the intrinsics of the data. **Cleaning:** The cleaning process was divided into two: part 1, and part 2. Part 1 contains the essential cleaning of the dataset, while part 2 objective was to prepare the data for it

to be in a tidy format in order to easily visualize the water balance (supply - use) and consumption (use / supply) of the countries and the economic activities without the need for further coding.

Part 1

When it came to start cleaning the raw data (*df_water*), I started by first selecting out unnecessary, redundant, and empty variables by creating a new dataframe. Empty variables (2) only contained missing values (NAs). The redundant variables described data in a slightly different format but in essence the same thing, thus, they were not providing meaningful information. Unnecessary variables contained only one unique value. After filtering out all these variables ($n=20$), the new dataset (*df*) contained the wanted variables ($n=8$). I checked for missing values. I renamed all variables into more intuitive names and changed some of them into snake format (*x_y*) as they were in space with backticks format (*`x y`*) which can cause issues. I renamed two values of distinct variables as well (a typo and a long name). I changed the structure of some variables with the mutate() function from character into factors. This with the aim of easier manipulation when plotting. One year was chosen from the year variable ($n=43$) based on three requirements: the year should have the most number of countries, observations, and be the closest to 2025. Resulted in filtering the year 2019 in a new data frame (*df_2019*) as it had 4 countries and observations less ($n=53$) ($n=1536$) than the year with most ($n=57$) ($n=1540$) and the closest to 2025. Following the aim of the RQ, it is relevant to have countries that have both values of water supply and use. Thus, an object was created grouping by country and summarizing which contained a list of the countries that had both water use and supply. The countries having both supply and use were filtered in new data frame (*df_final*). Lastly, the column containing year 2019 was removed as it was not useful anymore.

Part 2

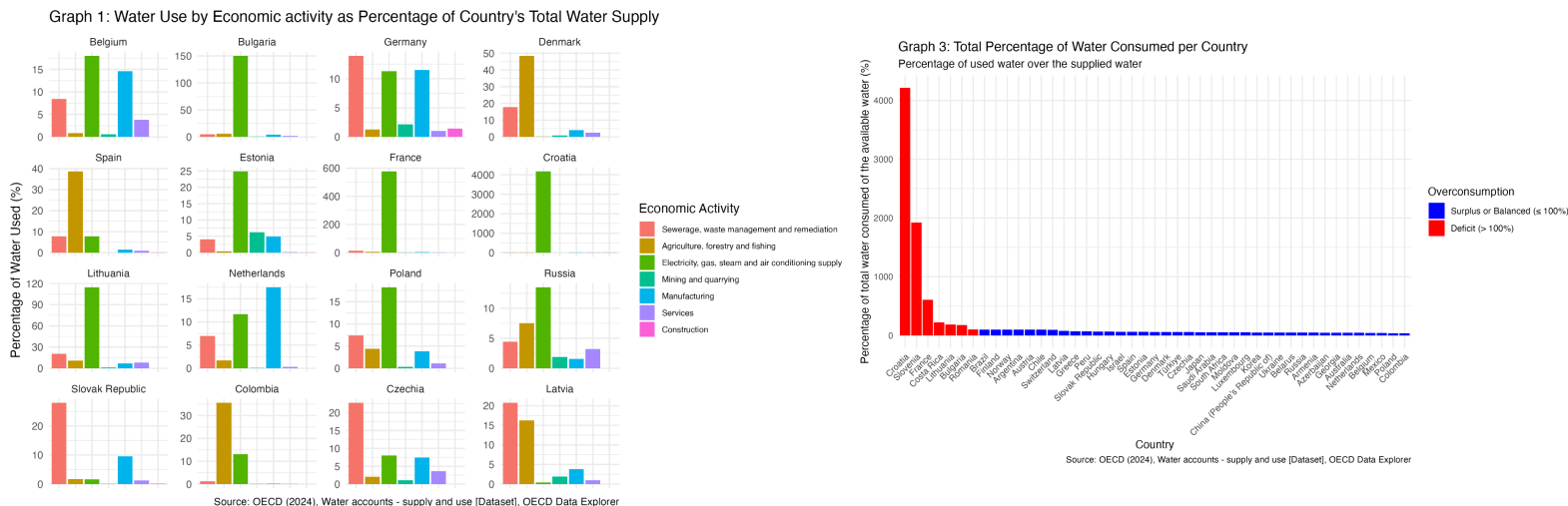
The cleaning continues with pivoting wider the variable "Supply and use" with the intention of having two separate columns with only use and only supply with their respective numerical values from the variable "Value in Millions". Next, the intention is to create a dataset (*df_all*) with all countries and all economic activities by grouping by country, economic activity and summarizing for a new variable measuring "Water balance" (supply minus use). This variable is important as it will describe if a sector used all the available water (balanced), if they overused it (deficit), or if they underused it (surplus). To be able to know how much percentage of the available water was used, a new variable "Water used percentage" was coded $((\text{use}/\text{supply}) * 100)$. A 100% would mean all water available was used, over 100% would describe how many times of the available water was overused, and below 100% would tell how much of the available water was used and unused. Therefore, to be able to see clearly in a visualization if water was overused or not, a new variable "Water overuse" was coded in a binomial form. TRUE meant the water was overused, and FALSE meant there was either a balance or underuse. The final steps of the cleaning are to create two datasets: one with all countries and only the value "Total - all activities" of economic activities variable (df_total); and another with only countries that have all 8 economic activities except for "Total - all activities" (df_econ). (*df_total*) helps in easier visualization of the worldwide trends for easier country comparisons, while (*df_econ*) aids in detailed accounts of economic sectors per country. As there were issues with computing operations for the use and supply information of the remaining 7 economic activities by summing up all supply values of all these activities; the "Total - all activities" values in supply were used as a reference of total water availability per country for all its economic sectors. Thus, the total supply row values from (*df_total*) was selected by country, and created a new variable "Country Total Supply". Then left join() by country incorporated the values in the dataframe. Lastly, a new variable "Sector use percentage" $((\text{use} / \text{Country_Total_Supply}) * 100)$ describing the percentage of water used per sector of the total available water in the country, was added to (*df_econ*).

4. Results

Question: Provide and discuss your preliminary results and visualizations.

The results did not need to be summarized as they are ready to just be plotted. I did three different visualizations. Graph 1, visualizes 16 countries with the percentages of water used by economic sector as a total of supplied water of the country. Overall, the electricity sector seems to be consuming the most amount of water relative to the availability, while the construction sector seems to be consuming the least. On the other hand, Graph 3, visualizes 47 countries with the percentage of total water consumption indicating the presence or absence of overconsumption. Most countries did not overconsume their water, except for 7 countries of which 2 (Slovenia and Croatia) overconsumed by more than 1000% of their total water supply. This suggests that either these countries found alternative ways to supply their water demand that were not annotated in the data, or these countries might be having unsustainable water consumption and run into water shortages. Additionally, although it was not included here due to space limits, Graph 2, visualizes 47 countries with numeric amounts of water balance describing the quantity of water deficit, balance, or surplus (in million cubic meters).

NOTE: See Graph 1, 2, and 3 in Rmarkdown or in their .png formats in the folder for better and detailed visualization. Additionally, Graph 1, in Rmarkdown has an interactive version as a shiny format. Run Rmarkdown to see more pleasing and appealing visualizations of all graphs.



5. References

Damkjaer, S. and Taylor, R. (2017). The Measurement of Water scarcity: Defining a Meaningful Indicator. *Ambio*, 46(5), pp.513–531. doi:<https://doi.org/10.1007/s13280-017-0912-z>.

OECD (2024). *OECD Data Explorer*. [online] Oecd.org. Available at: <https://data-explorer.oecd.org/vis?lc=en&df>.